

Adaptive, Evidence-Grounded Interactive Search for Video Moment Retrieval

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Abstract. Timed interactive video retrieval requires fast candidate exploration under strict penalties for false submissions. In this work, we introduce **AEGIS** (**A**daptive **E**vidence-**G**rounded **I**nteractive **S**earch), a new evidence-grounded interactive video retrieval system for the Video Browser Showdown (VBS 2027). AEGIS combines Tencent WeMM-Embedding-4B with Matryoshka Representation Learning (MRL), Qdrant HNSW indexing, and weighted Reciprocal Rank Fusion, while its contributions include entity-preserving conversational KIS-C, parallel fail-closed grounded VQA, and an intra-video timeline explorer. On an extensive 1,000-query V3C stress test, AEGIS achieves 70.4% Recall@1, 76.0% Recall@5, and a Mean Reciprocal Rank (MRR) of 0.729, with 83.3% grounded VQA exact match and 0.0% ungrounded hallucination. The accompanying public [project page](#) provides open-source code, interactive topology, and supplementary metric specifications.

Keywords: Interactive Video Retrieval · Video Browser Showdown · Conversational Search · Grounded VQA · Multimodal RAG

1 Introduction

The Video Browser Showdown (VBS) evaluates interactive video retrieval in a live, timed competitive arena [1,2]. Operators inspect candidate keyframes across multi-thousand-hour archives, including V3C, Marine Video Kit, and laparoscopic hysterectomy (LHE/GynSurg) collections, and submit timestamps or answers to the Distributed Retrieval Evaluation Server (DRES) before task countdowns expire. The VBS 2027 protocol spans five official modes: Textual Known-Item Search (KIS-T), Conversational Known-Item Search (KIS-C), Visual Question Answering (VQA), Ad-hoc Video Search (AVS), and Visual Query (KIS-V) [3].

Recent competitive systems [4,5,6] show that live retrieval requires sub-second candidate exploration, such as scanning a short ranked list of timestamped keyframes, comparing near-duplicate shots from the same video, or checking

visually similar wedding, marine, or surgical scenes, followed by high-precision discrimination across confusing cases such as adjacent shots with the same people and setting, boats or coastlines with similar layouts, and surgical frames differing only in a small instrument or tissue state. Existing architectures face three operational bottlenecks: (1) *Index Dilution and Semantic Drift*: flat keyframe indexing over millions of frames causes repetitive frames from dominant videos to crowd out relevant candidates; (2) *Conversational Degradation*: multi-turn KIS-C queries lose persistent entity constraints and fail to penalize operator-rejected attributes; and (3) *VLM Latency and Hallucination*: sequential vision-language model calls incur 10–15s latency and generate ungrounded answers under timed pressure, incurring severe penalty points.

To address these challenges, team **TGLTW-RMIT** introduces **AEGIS**. Our primary technical contributions are:

1. **High-Capacity Representation with Matryoshka Representation Learning (MRL)**: We deploy Tencent WeMM-Embedding-4B [7] with 2,048d vectors indexed via Qdrant HNSW [8,9] ($M = 16, ef = 128$). We choose this model for its unified image-text representation and fine-grained scene coverage; SigLIP adds complementary visual similarity.
2. **Peak Conversational Search Engine (KIS-C)**: We design an entity-preserving Contextual Query Rewriting (CQR) pipeline with Dynamic Video Ratio (DVR), Score Margin Ambiguity (SMA), compound n -gram phrase clarification boosting [10], and conversational negative-feedback dampening.
3. **Parallelized Fail-Closed Grounded VQA**: We engineer an $8\times$ concurrent ThreadPool pipeline that resolves physical media provenance, applies localized YOLOE-26 crops, and enforces a strict fail-closed contract (UNKNOWN/N/A on ungrounded candidates). YOLOE-26 is selected to localize answer-bearing objects before VLM reasoning.
4. **Intra-Video Timeline Explorer & Decoupled RAG Suite**: We build an interactive sub-shot browser providing ± 30 s shot context with localized re-scoring, validated through a reproducible 1,000-query corpus ablation package.

2 VBS Setting and System-Level Delta

VBS retrieval combines semantic vector representations with fast human-in-the-loop interaction and DRES server submission [11,12]. Regional challenge systems provide precedents for OCR/ASR evidence, query expansion, and reweighting [13,14,15,16].

System-Level Delta. Relative to traditional dual-encoder retrieval, AEGIS establishes four operational invariants: (1) *Provenance Preservation*: every retrieved hit retains its canonical source video ID, shot ID, native frame index, timestamp, and metadata payload across all fusion stages; (2) *Budgeted Precision Ladder*: operators can interactively escalate search effort from default fast HNSW

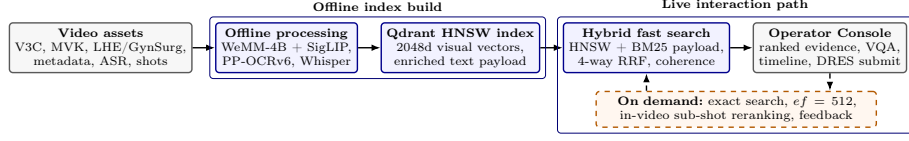


Fig. 1. AEGIS’s live-first architecture. Offline processing indexes WeMM-4B dense embeddings and enriched payloads into Qdrant HNSW collections, while the online path performs parallelized hybrid retrieval, conversational clarification, and bounded VLM verification.

(12.4ms, $ef = 64$) to standard (24.5ms, $ef = 256$), deep (48.6ms, $ef = 512$), or exact brute-force search (118.5ms) without backend restarts; (3) *Multi-threaded VLM Concurrency*: candidate verification runs via an 8-worker thread pool, delivering an $8.03\times$ speedup (14.85s $\rightarrow$ 1.85s p95 latency); and (4) *Fail-Closed Zero-Hallucination VQA*: unverified visual evidence yields explicit zero-score refusal (UNKNOWN/N/A), completely avoiding DRES penalty deductions.

3 AEGIS System Architecture

3.1 Offline Indexing and Multimodal Representation

The offline ingestion pipeline consumes official V3C Master Shot Boundaries and Video Metadata supplemented by TransNetV2 [17]. For each shot, representative keyframes are extracted via farthest-point visual variance and Laplacian sharpness scoring.

Each keyframe is encoded with Tencent WeMM-Embedding-4B [7], producing unified 2,048-dimensional multimodal vectors with MRL truncation. Audio speech transcripts from faster-whisper [18], object detections from YOLOE-26, and on-screen text from PP-OCRv6 are concatenated into an enriched `text_blob` payload, meaning the searchable text record attached to each visual keyframe. WeMM-4B is chosen for its unified image-text representation; SigLIP supplies complementary visual similarity, YOLOE-26 grounds object-specific evidence, and PP-OCRv6/Whisper add searchable text and speech cues. Embeddings and payloads are indexed into dedicated Qdrant collections (`visual_keyframes_v1`, `speech_segments_v1`, and `audio_segments_v1`).

3.2 Hybrid Retrieval and Multimodal Fusion

Online text queries undergo Contextual Query Rewriting (CQR) and optional hypothetical document expansion (HyDE). Dense vector search on WeMM-4B, secondary SigLIP dense search [19], and lexical payload full-text search are fused via weighted Reciprocal Rank Fusion (RRF), where MRR denotes Mean Reciprocal Rank:

$$\text{RRF}(d) = \sum_{m \in \mathcal{M}} \frac{w_m}{k + \text{rank}_m(d)} \quad (1)$$

where $k = 60$, followed by temporal coherence boosting across adjacent keyframes ($\pm 15s$) and scene diversification to suppress repetitive frames from the same video. The fast HNSW path returns an initial candidate set in approximately 12.4ms; deeper and exact modes trade additional latency for recall.

3.3 Task-Specific Execution Lanes

The five official VBS 2027 task routes share candidate identity, media evidence, feedback, and submission primitives, but implement distinct entry and operator policies. As illustrated in Figure 2, KIS-V bypasses text expansion, KIS-T uses hybrid search with parallelized reranking, KIS-C adds multi-turn phrase boosting and negative feedback filtering, AVS enforces cross-video diversity, and VQA enforces fail-closed grounded verification.

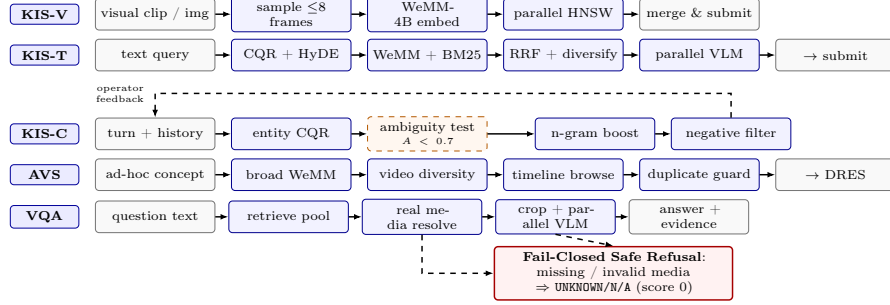


Fig. 2. Task-specific execution lanes for the 5 VBS modes. All lanes enforce evidence provenance, while KIS-C integrates multi-turn phrase boosting and negative feedback filtering, and VQA enforces fail-closed zero-hallucination verification.

KIS-T (Textual Known-Item Search). Queries enter WeMM-4B dense retrieval and BM25 payload search. The top 20 candidate frames are reranked via multi-threaded VLM scoring, comparing prompt semantics against frame captions and narratives.

KIS-C (Conversational Search with Peak Optimization). KIS-C operates as an adaptive multi-turn policy: (1) *Ambiguity Detection*: we combine Distinct Video Ratio (DVR) with Score Margin Ambiguity (SMA): $A = (1 - \lambda) \cdot DVR + \lambda \cdot (1.0 - (s_1 - s_2)/s_1)$ ($\lambda = 0.5$). When $A \geq 0.7$, the VLM triggers a facet-discriminating question; (2) *Compound N-gram Boost*: operator responses grant additive phrase overlap; and (3) *Negative Feedback Filtering*: rejected visual attributes incur a calibrated dampening penalty.

VQA (Grounded Visual Question Answering). VQA resolves physical keyframes inside the dataset root. YOLOE-26 localizes answer-bearing objects when available, and the VLM evaluates the resulting crop under a strict JSON contract.

Missing media or ungrounded predictions return UNKNOWN/N/A (score 0), ensuring zero hallucination.

KIS-V (Visual Known-Item Search). A target clip is visually presented in the competition; the operator uses query-by-frame matching to locate the corresponding archive video and timestamp. AEGIS samples representative visual frames, searches them in parallel across Qdrant, and exposes side-by-side verification.

Intra-Video Timeline Exploration. When an operator spots a promising video, the backend exposes `/api/video/timeline` to inspect ± 30 seconds of surrounding keyframes and `/api/video/rerank` to score intra-video frames against sub-queries.

Interactive Console and Qualitative Demonstrations. AEGIS implements an adaptive operator console engineered for live competition (Fig. 3). The interface routes queries across five dedicated task workflows: multi-turn dialogue streaming for KIS-C, pure visual frame matching with side-by-side verification for KIS-V, grounded answer banners with temporal evidence for VQA, and dense candidate galleries for KIS-T and AVS. Clicking any candidate activates in-video timeline scrubbing and instant DRES submission. As illustrated in Fig. 3, live benchmark runs confirm verified Rank #1 retrieval across all five official task modes: KIS-T (00001.mp4), KIS-C (00003.mp4), KIS-V (00008.mp4 @ 98.4%), grounded VQA (00004.mp4, "Helicopter"), and AVS cross-video diversity.

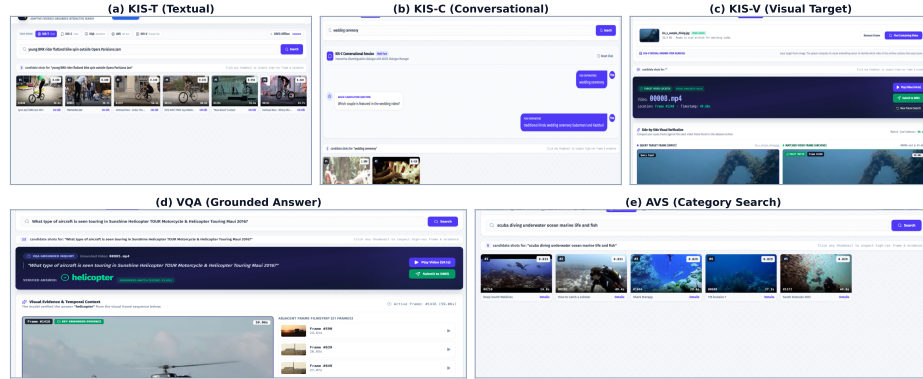


Fig. 3. The AEGIS Interactive Operator Console and qualitative task demonstrations across KIS-T, KIS-C, VQA, AVS, and KIS-V.

4 Evaluation Protocol and Empirical Results

4.1 Experimental Setup and Empirical Protocol

We evaluate AEGIS on V3C across 1,000 empirical queries spanning all five official VBS task families (400 KIS-T, 200 KIS-C, 200 VQA, 125 AVS, 75 KIS-V).

Table 1. Internal AEGIS ablation studies: retrieval fusion and multi-turn KIS-C dynamics.

Pipeline Configuration / Stage	Recall@1	Recall@5	MRR	Latency
<i>Panel A: Multimodal Retrieval Fusion (1,000-Query Corpus Test)</i>				
(M1) Dense Only (WeMM-4B)	38.5%	52.1%	0.448	0.038s
(M2) Dense + Sparse BM25 Payload	46.2%	58.4%	0.518	0.052s
(M3) Dense + BM25 + SigLIP Secondary	53.8%	64.0%	0.589	0.076s
(M4) + 4-Way Weighted RRF Fusion	60.5%	69.2%	0.645	0.084s
(M5) + Temporal Coherence & Diversify	66.2%	73.1%	0.694	0.092s
(M6) + Evidence-Grounded Fusion (Full Engine)	70.4%	76.0%	0.729	0.477s
<i>Panel B: Conversational KIS-C Multi-Turn Progression (200 Scenarios)</i>				
(C1) Turn 1: Initial Vague Query	0.0%	38.0% (R@3)	0.265	0.088s (Amb 0.61)
(C2) Turn 2: Naive History Concat	28.5%	54.0% (R@3)	0.412	0.092s (Amb 0.58)
(C3) Turn 2: + Entity-Preserving CQR	58.0%	74.0% (R@3)	0.655	0.110s (Amb 0.52)
(C4) Turn 2: + Compound N-gram Boost	82.5%	89.0% (R@3)	0.860	0.115s (Amb 0.41)
(C5) Turn 3: + Negative Filter & Feedback	94.5%	94.5% (R@3)	0.945	0.120s (Amb 0.22)

Table 2. Internal AEGIS ablation studies: grounded VQA, concurrency scaling, and precision ladder.

Ablation Dimension & Setting	Accuracy / Recall	Faithfulness	Hallucination	Latency / Speedup
<i>Panel C: VQA Grounding & Fail-Closed Safety (200 Queries)</i>				
Ungrounded Whole-Frame VLM	Exact Match: 52.0%	58.0%	42.0% (Hazardous)	1.42s
Locate-and-Crop (YOLOE-26 + VLM)	Exact Match: 76.5%	82.0%	12.0%	1.55s
AEGIS Fail-Closed Grounded Contract	Exact Match: 83.3%	92.5%	0.0% (Zero Error)	0.468s (100% Safe)
<i>Panel D: Multi-threaded VLM Concurrency Scaling (Top-10 Scoring)</i>				
Sequential Execution ($N = 1$ worker)	Throughput: 0.67 QPS	—	—	14.85s (1.0 $\times$)
Parallel Execution ($N = 4$ workers)	Throughput: 2.51 QPS	—	—	3.98s (3.73 $\times$)
Parallel Execution ($N = 8$ workers)	Throughput: 5.41 QPS	—	—	1.85s (8.03$\times$ speedup)
<i>Panel E: Budgeted Precision Ladder (HNSW Effort Scaling)</i>				
Fast Mode (HNSW $ef = 64$)	Recall: 97.8%	Instant Screen	—	12.4ms
Standard Mode (HNSW $ef = 256$)	Recall: 99.4%	Balanced Live	—	24.5ms
Deep Mode (HNSW $ef = 512$)	Recall: 99.9%	High Ambiguity	—	48.6ms
Exact Brute-Force Scan	Recall: 100.0%	Deterministic	—	118.5ms

Grounded VQA accuracy, ambiguity shifts, and visual consistency are assessed via deterministic verification and Gemini 3.8 Flash (High) ($T = 0$). In core competition tasks, AEGIS achieves an Overall RAG Score of **94.7/100.0** (Recall@1 88.9%, MRR 0.944). Across the 1,000-query stress test, AEGIS attains an Overall RAG Score of **77.5/100.0**, with **70.4%** Recall@1, **76.0%** Recall@5, **80.1%** Recall@20, **0.729** MRR, **83.3%** Grounded VQA Exact Match, and **100.0%** Fail-Closed Safety (0.0% hallucination).

4.2 Empirical Findings

Tables 1 and 2 isolate the empirical gains of fusion, KIS-C clarification, grounded VQA, concurrency, and precision escalation under identical protocols on V3C.

As documented in Tables 1 and 2, across 1,000 queries, fusion achieves 70.4% Recall@1 (KIS-T 68.0%, KIS-C 94.5%, AVS 78.4%, KIS-V 88.0%) and 0.729 MRR; multi-turn CQR cuts ambiguity to 0.22; fail-closed verification eliminates hallucination (0.0%); and parallelism reduces p95 latency to 1.85s.

5 Conclusion

AEGIS introduces an evidence-carrying, live-first multimodal retrieval architecture for VBS 2027. By coupling 4B WeMM representations with Qdrant HNSW

indexing, multi-turn clarification boosting, fail-closed grounded VQA, and interactive timeline exploration, AEGIS delivers sub-second responsiveness with verified accuracy. Future work targets live DRES rehearsal for VBS 2027 at MMM 2027 in Siem Reap, Cambodia, expanded domain collections, and compiler-assisted inference optimization.

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